Experiment No 1.1:

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Branch: BE CSE specialization Cyber Security

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1. Aim/Overview of the practical:

Simulate / Understand the working of following (i) IP Address. (ii) Cisco IOS. (iii) Straight Cable & Cross Cable, RJ45 (iv) Layer 2 Switch. (v) Router

2. Software/Tools Required:

- Cisco Packet Tracer
- Windows Operating System

3. Theory:

A) IP Address

What is an IP Address?

An IP (Internet Protocol) Address is a unique logical address assigned to every device connected to a network.

Think of it like this:

- Every house has a unique postal address.
- Every device on a network has a unique IP address.

Without an IP address, data would not know where to go.

Example

Suppose:

- Your laptop → 192.168.1.10
- Friend's laptop → 192.168.1.20

When you send a file, the network looks at the destination IP (192.168.1.20) and delivers it to your friend's laptop.

IPv4 vs IPv6

Feature	IPv4	IPv6
Address Length	32 bits	128 bits
Format	Decimal	Hexadecimal
Example	192.168.1.10	2001:db8::1
Number of Addresses	~4.3 Billion	Almost Unlimited
Usage	Most common	Increasing rapidly

Classes of IPv4

classs	Range	Default Subnet Mask	Used For
A	1–126	255.0.0.0	Large organizations
B	128–191	255.255.0.0	Medium organizations
C	192–223	255.255.255.0	Small offices, homes
D	224–239	N/A	Multicasting
E	240–255	N/A	Research & Testing

What is a Subnet Mask?

A Subnet Mask tells a computer:

- Which part of the IP address is the Network ID
- Which part is the Host ID

Example:

IP Address

192.168.10.25

Subnet Mask

255.255.255.0

B) Cisco IOS (Internetwork Operating System)

Cisco IOS is the operating system developed by Cisco Systems for its networking devices such as routers and switches. It controls the hardware of the device and provides the command-line interface (CLI) through which network administrators configure, monitor, and manage the network.

In simple words, Cisco IOS is to a Router/Switch what Windows is to a Laptop.

Why Do We Need Cisco IOS? When a new Cisco router is purchased, it does not initially know which IP address to use, which interface should be enabled, which network is connected to which port, where to send incoming packets, or whether to allow/block specific traffic. These decisions are made by the administrator using Cisco IOS. Without Cisco IOS, a router cannot perform its networking functions.

What Can Cisco IOS Do?

- Assign IP addresses to interfaces
- Enable or disable network ports
- Configure routing between different networks
- Create Virtual LANs (VLANs) on switches
- Set passwords for security
- Monitor network traffic
- Test connectivity using commands like ping
- View the routing table
- Save and restore device configurations

Mode	**Prompt**	**Command to Enter**	**Purpose**
User EXEC	Router>	Automatically after boot	Basic monitoring commands
Privileged EXEC	Router#	enable	Administrative commands, verification
Global Configuration	Router(config)#	Configure terminal	Configure the router
Return to Previous Mode	Router(config)# → Router#	end or Ctrl + Z	Exit configuration mode

C) Straight Cable and Cross Cable (RJ-45)

Why Different Cables? For successful communication, the Transmit (Tx) pins of one device must connect to the Receive (Rx) pins of the other device.

- Correct: Tx → Rx and Rx → Tx
- Incorrect: Tx → Tx and Rx → Rx

Types of Devices

1. End Devices (MDI): These devices generate or consume data — PC, Laptop, Server, Router
2. Intermediate Devices (MDI-X): These devices forward data between devices — Switch, Hub

Wiring Standards (RJ45 – 8 Pins)

Pin	T568A	T568B
1	White-Green	White-Orange
2	Green	Orange
3	White-Orange	White-Green
4	Blue	Blue
5	White-Blue	White-Blue
6	Orange	Green
7	White-Brown	White-Brown
8	Brown	Brown

Feature	Straight Cable	Cross Cable
Definition	Both ends use the same wiring standard (T568A-T568A or T568B-T568B)	One end uses T568A and the other uses T568B
Internal Wiring	No crossing of wires	Transmit and receive pairs are crossed
Purpose	Connects different types of devices	Connects similar types of devices
Typical Connections	PC-Switch, Switch-Router, PC-Hub, Router-Hub	PC-PC, Switch-Switch, Router-Router, Hub-Hub, PC-Router (older devices)
Tx/Rx Mapping	Relies on one device internally handling Tx/Rx	Cable itself swaps Tx and Rx lines
Ease of Identification	Same color sequence at both ends	Different color sequence at each end
Modern Networks	Commonly used	Less common due to Auto-MDIX

D) Layer 2 Switch

Theory of a Layer-2 Switch A Layer-2 Switch is an intelligent networking device that operates at the Data Link Layer (Layer 2) of the OSI Model. Its primary function is to connect multiple devices within the same Local Area Network (LAN) and efficiently forward data frames from the sender to the intended receiver. Unlike a hub, which broadcasts data to every connected device, a switch learns the MAC addresses of connected devices and forwards frames only to the correct destination, thereby reducing unnecessary network traffic and improving network performance.

Theory Points

1. **Operates at the Data Link Layer (Layer 2) —** A Layer-2 Switch works at the Data Link Layer of the OSI Model. It forwards Frames (not packets) using MAC Addresses instead of IP Addresses.
2. **Uses MAC Addresses for Communication —** Every network device has a unique MAC (Media Access Control) Address. The switch reads the Destination MAC Address from the incoming Ethernet frame and decides where to forward it.
3. **Builds a MAC Address Table (CAM Table) —** Whenever a frame enters the switch, it records the Source MAC Address and the port on which the frame arrived. This information is stored in a table called the CAM (Content Addressable Memory) Table or MAC Address Table. As more devices communicate, the table gradually grows.
4. **Reduces Network Traffic —** Since frames are delivered only to the required destination, unnecessary traffic is avoided. This improves bandwidth utilization and increases the overall efficiency of the network.
5. **Supports Full-Duplex Communication —** Modern switches allow devices to transmit and receive data simultaneously using Full-Duplex Mode. This eliminates collisions and doubles the effective communication capacity.

E) Router

What is a Router? A Router is a networking device that operates at the Network Layer (Layer 3) of the OSI Model. Its primary function is to connect two or more different networks and intelligently forward data packets from the source network to the destination network.

Characteristics of a Router

1. A Router operates at the Network Layer (Layer 3) of the OSI Model.

2. It forwards Packets, not Frames or Bits.
3. It makes forwarding decisions using IP Addresses.
4. It connects two or more different networks.
5. It maintains a Routing Table that contains information about available networks and the paths to reach them.

3. Steps for experiment/practical:

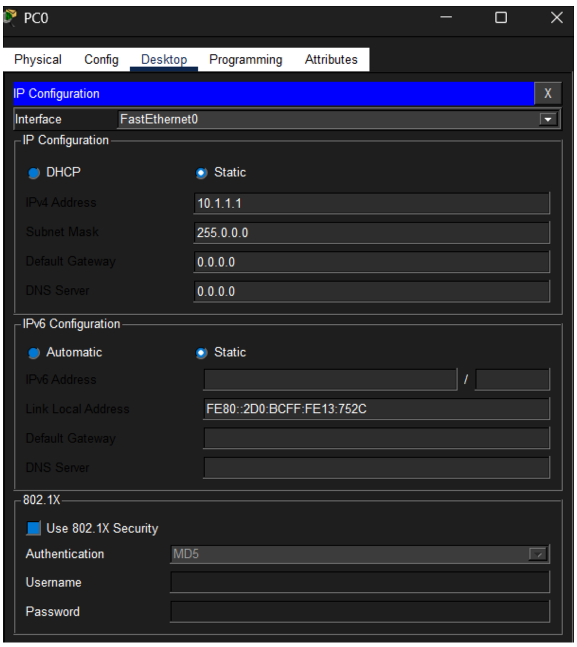
A) IP Address:

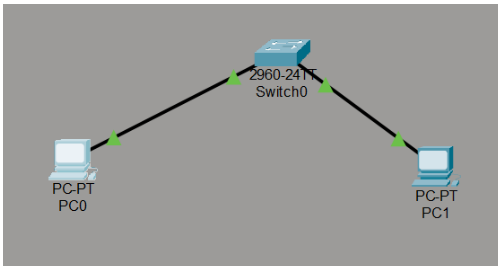
Steps

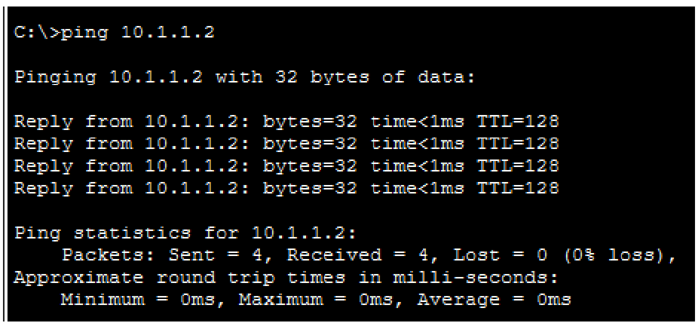
1. Open Cisco Packet Tracer.
2. Place two PCs.
3. Connect both PCs using a switch.
4. Click on PC0 → Desktop → IP Configuration.
5. Assign:
   - IP Address
   - Subnet Mask
6. Repeat for PC1.
7. Open the Command Prompt.
8. Execute:

ping 192.168.1.20

If the reply is received, communication has been established successfully.







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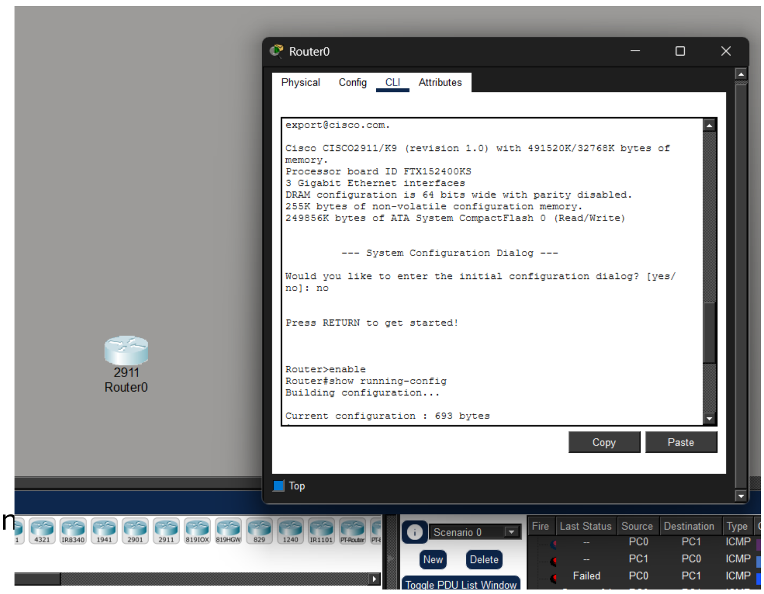
B) Cisco IOS:

Procedure / Steps

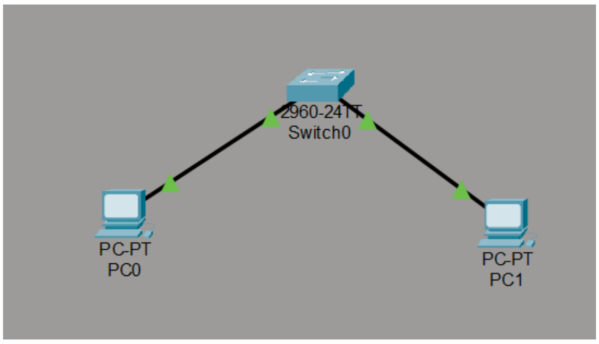
1. Open **Cisco Packet Tracer**.
2. Drag a **Router (Cisco 2911)** into the workspace.
3. Click on the router and open the **CLI** tab.
4. Press **Enter** to start the Cisco IOS Command-Line Interface.
5. Observe the **User EXEC Mode** prompt:
   Router>
6. Execute the following commands in **User EXEC Mode**:
   - ?
   - show ?
   - show clock
   - show version
   - ping 127.0.0.1
   - traceroute 127.0.0.1
7. Enter **Privileged EXEC Mode** by typing:
   - enable
8. Execute the following commands in **Privileged EXEC Mode**:
   - show running-config
   - show startup-config
   - show ip interface brief
   - show history
9. Enter **Global Configuration Mode** by typing:
   - configure terminal
10. Execute the following configuration commands:
    - hostname R1
    - banner motd #Unauthorized Access Prohibited#
    - enable secret cisco123
    - service password-encryption
11. Exit the configuration mode using:
    - End

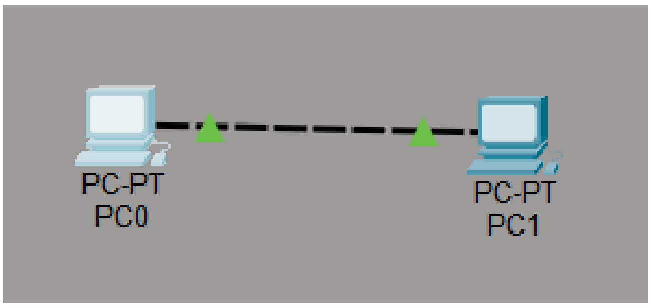


C) Straight Cable & Cross Cable

Steps:

1. Open Cisco Packet Tracer.
2. Select Connections (Lightning icon).
3. Choose Copper Straight-Through cable.
4. Connect PC → Switch.
5. Choose Copper Cross-Over cable.
6. Connect PC → PC or Switch → Switch.
7. Observe the successful connection between the devices.

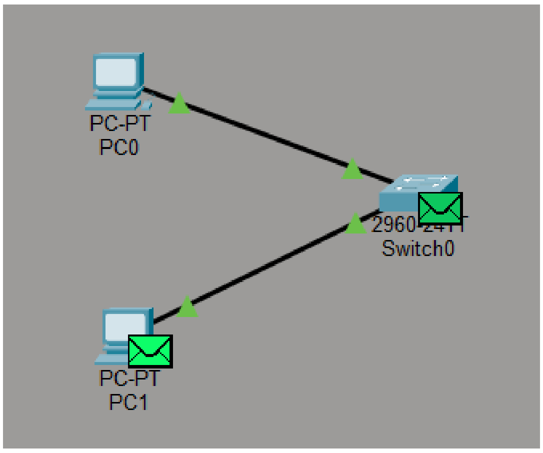




D) Layer-2 Switch

Procedure

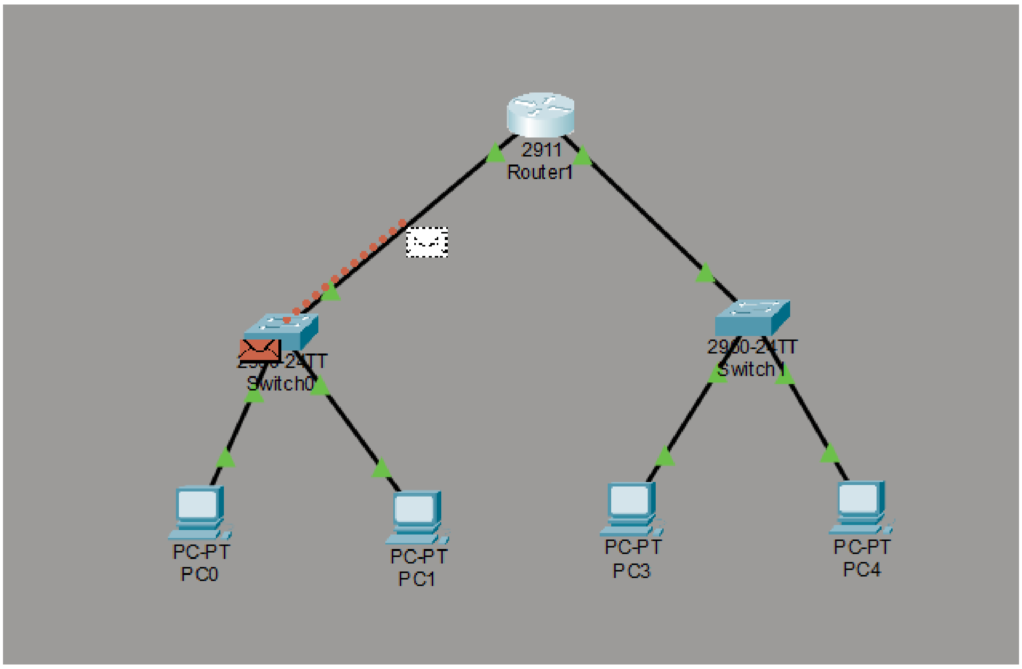
1. Open **Cisco Packet Tracer**.
2. Place **one 2960 Switch** and **two PCs** in the workspace.
3. Connect both PCs to the switch using **Copper Straight-Through** cables.
4. Assign IP addresses to both PCs in the same network.
5. Open the Command Prompt on one PC and **ping** the other PC.
6. Verify that the communication is successful.



D) Router

Procedure

1. Open **Cisco Packet Tracer**.
2. Place **one Router**, **two Switches**, and **two PCs** in the workspace.
3. Connect the PCs to the switches and the switches to the router using **Copper Straight-Through** cables.
4. Assign IP addresses to the PCs and router interfaces.
5. Configure the PCs' **Default Gateway** with the router's IP address.
6. Verify connectivity using the ping command.
7. Display the routing table using:
   show ip route



4.Result

The working of **IP Address, Cisco IOS, Straight Cable & Cross Cable, Layer-2 Switch, and Router** was studied successfully. IP addressing was configured, Cisco IOS commands were executed, and communication between different networks was established successfully using a router.

5.Learning Outcomes

- Understood the concept and configuration of **IP addresses** and subnet masks.
- Learned to use basic **Cisco IOS CLI** commands for router configuration.
- Identified the use of **Straight-Through** and **Cross-Over** cables in networking.
- Understood the working of a **Layer-2 Switch** and MAC address-based forwarding.
- Configured a **Router** to connect two different networks and verified connectivity using the ping command.

Evaluation Grid (To be created as per the SOP and Assessment guidelines by the faculty):

Sr. No.	Parameters	Marks Obtained	Maximum Marks
1.	Worksheet completion		08
2.	Post Lab Quiz Result/Viva-Voce		10
3.	Conduct		12
	Total		**30**